EL6 Worked Examples: Solving Exponential and Logarithmic Equations

Recording companion for the EL6 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

What is wrapped around the power or the logarithm, and what do I have to undo before I switch forms? Undo the outside first: add or subtract the number that is outside, then divide by any multiplier. Once the power or the logarithm is alone, switch forms: log base b of x equals y means b to the y equals x. Then finish solving for the variable and leave the answer exact.

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Example 1: First Move 1

$$5^{x-3}=8$$

Question

What is the best first move to solve for x?

- **A.** Subtract 3 from both sides.
- **B.** Divide both sides by 5.
- **C.** Switch to logarithmic form: $x-3=\log_5 8$
- **D.** Switch to logarithmic form: $x-3=\log_8 5$

Answer and reasoning

The power 5^{x-3} is already alone on the left, so you can switch forms right away. The base stays the base: $5^{x-3}=8$ means $x-3=\log_5 8$. The 3 is inside the exponent, so you cannot subtract it from the whole side, and dividing by 5 does not undo a power. **The first move is to switch to $x-3=\log_5 8$.**

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Example 2: First Move 2

$$2^x - 6 = 9$$

Question

What is the best first move to solve for x?

- **A.** Add 6 to both sides.
- **B.** Switch to logarithmic form: $x - 6 = \log_2 9$
- **C.** Divide both sides by 2.
- **D.** Subtract 9 from both sides.

Answer and reasoning

The 6 is outside the power, so the power is not alone yet. Add 6 to both sides first: $2^x = 15$. Writing $x - 6 = \log_2 9$ moves the 6 into the exponent, but the exponent is only x.
The first move is to add 6 to both sides, giving $2^x = 15$.

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Example 3: First Move 3

$$4\log_3(2x)=8$$

Question

What is the best first move to solve for x?

- **A.** Switch to exponential form: $3^8=2x$
- **B.** Subtract 4 from both sides.
- **C.** Divide both sides by 2.
- **D.** Divide both sides by 4.

Answer and reasoning

The 4 multiplies the logarithm, so the logarithm is not alone yet. Divide both sides by 4 first: $\log_3(2x)=2$. Only then switch forms: $2x=3^2=9$. Writing $3^8=2x$ skips the division and leaves out the 4. **The first move is to divide both sides by 4, giving $\log_3(2x)=2$.**

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Example 4: First Move 4

$$7+\log_2(5x)=10$$

Question

What is the best first move to solve for x?

- **A.** Switch to exponential form: $2^{10}=5x$
- **B.** Subtract 7 from both sides.
- **C.** Divide both sides by 7.
- **D.** Divide both sides by 5.

Answer and reasoning

The 7 is added to the logarithm, so subtract it first: $\log_2(5x)=3$. Now the logarithm is alone, and switching forms gives $5x=2^3=8$. Writing $2^{10}=5x$ ignores the 7. **The first move is to subtract 7 from both sides, giving $\log_2(5x)=3$.**

Example 5: Exponential 1

$$3^{x-4}=7$$

Question

Solve for x.

A. $x=\log_3(7)-4$
B. $x=\log_3(11)$
C. $x=\log_7(3)+4$
D. $x=\log_3(7)+4$

Answer and reasoning

The power is already alone, so switch forms with base 3: $x-4=\log_3 7$. Then add 4 to both sides: $x=\log_3(7)+4$. A quick check: $\log_3 7$ is about 1.77, so x is about 5.77, and $3^{1.77}$ is about 7. **The solution is $x=\log_3(7)+4$.**

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Example 6: Exponential 2

$$2^x - 5 = 9$$

Question

Solve for x.

A. $x = \log_2(14)$
B. $x = \log_2(9) + 5$
C. $x = \log_2(4)$
D. $x = \log_{14}(2)$

Answer and reasoning

The 5 is outside the power, so add it first: $2^x = 14$. Now switch forms: $x = \log_2(14)$. Check: $2^{\log_2 14} = 14$, and $14 - 5 = 9$. The answer $\log_2(9) + 5$ moves the 5 into the exponent, and $\log_2(4)$ subtracts 5 when it should add. **The solution is $x = \log_2(14)$.**

Example 7: Exponential 3

$$4^{k-3}=6$$

Question

Solve for k.

A. $k=\log_4(6)-3$
B. $k=\log_4(6)+3$
C. $k=\log_6(4)+3$
D. $k=\log_4(9)$

Answer and reasoning

The variable is k this time, and that changes nothing. The power is alone, so switch forms with base 4: $k-3=\log_4 6$. Add 3 to both sides: $k=\log_4(6)+3$. The answer $\log_4(9)$ adds 3 inside the logarithm, but the 3 was never with the 6. **The solution is $k=\log_4(6)+3$.**

Example 8: Exponential 4

$$3^m - 8 = 4$$

Question

Solve for m.

A. $m = \log_3(4) + 8$
B. $m = \log_{12}(3)$
C. $m = \log_3(12)$
D. $m = \log_3(-4)$

Answer and reasoning

Add 8 to both sides first: $3^m = 12$. Then switch forms: $m = \log_3(12)$. Subtracting 8 instead gives $\log_3(-4)$, a logarithm of a negative number. That is not defined, which is a sign something went wrong. **The solution is $m = \log_3(12)$.**

Example 9: Logarithmic 1

$$3\log_2(5x)=9$$

Question

Solve for x.

A. $x=8/5$
B. $x=512/5$
C. $x=6/5$
D. $x=5/8$

Answer and reasoning

Divide by 3 first: $\log_2(5x)=3$. Switch forms: $5x=2^3=8$. Divide by 5: $x=8/5$. Check: $\log_2 8=3$, and $3\cdot 3=9$. The answer $512/5$ is $(2^9)/5$, which skips dividing by 3. The answer $6/5$ uses $2\cdot 3$ instead of 2^3. **The solution is $x=8/5$.**

Example 10: Logarithmic 2

$$6+\log_3(4x)=8$$

Question

Solve for x.

A. $x=3/2$
B. $x=4/9$
C. $x=9/4$
D. $x=6561/4$

Answer and reasoning

Subtract 6 first: $\log_3(4x)=2$. Switch forms: $4x=3^2=9$. Divide by 4: $x=9/4$. The answer $6561/4$ is $(3^8)/4$, which ignores the 6. The answer $3/2$ uses $3\cdot 2$ instead of 3^2, and $4/9$ divides the wrong way. **The solution is $x=9/4$.**

Example 11: Logarithmic 3

$$2\log_5(3w)=4$$

Question

Solve for w.

A. $w=625/3$
B. $w=75$
C. $w=10/3$
D. $w=25/3$

Answer and reasoning

Divide by 2 first: $\log_5(3w)=2$. Switch forms: $3w=5^2=25$. Divide by 3: $w=25/3$. The answer $625/3$ is $(5^4)/3$, which skips dividing by 2. The answer 75 multiplies by 3 at the end instead of dividing. **The solution is $w=25/3$.**

Example 12: Logarithmic 4

$$9+\log_4(7n)=12$$

Question

Solve for n.

A. $n=64/7$
B. $n=12/7$
C. $n=(4^{12})/7$
D. $n=7/64$

Answer and reasoning

Subtract 9 first: $\log_4(7n)=3$. Switch forms: $7n=4^3=64$. Divide by 7: $n=64/7$. The answer $(4^{12})/7$ switches forms before subtracting the 9, and $12/7$ uses $4\cdot 3$ instead of 4^3. **The solution is $n=64/7$.**

Example 13: Slip 1

$$2^{x-3}=5$$
$$x=\log_2(5)-3$$

Question

A student wrote these lines. What is the mistake?

- **A.** The base of the logarithm should be 5, not 2.
- **B.** The student should have divided 5 by 2 first.
- **C.** The last step should add 3, not subtract it.
- **D.** There is no mistake.

Answer and reasoning

Switching forms gives $x-3=\log_2 5$. To get x alone, add 3 to both sides. The student subtracted instead. **The correct solution is $x=\log_2(5)+3$.**

Example 14: Slip 2

$$4^x - 7 = 9$$
$$x - 7 = \log_4 9$$

Question

A student wrote these lines. What is the mistake?

A. The 7 is outside the power. Add it before switching forms.
B. The base of the logarithm should be 9, not 4.
C. Both sides should be divided by 4 first.
D. There is no mistake.

Answer and reasoning

The exponent is only x. The 7 is outside the power, so add it first: $4^x = 16$. Then $x = \log_4(16)$, which simplifies to 2, because $4^2 = 16$. Check: $4^2 - 7 = 9$. **The correct solution is $x = \log_4(16) = 2$.**

Example 15: Slip 3

$$5\log_3(2x)=10$$
$$2x=3^{10}$$

Question

A student wrote these lines. What is the mistake?

A. The 3 and the 10 should trade places.
B. The student switched forms before dividing by 5.
C. The 2 should have been subtracted, not divided.
D. There is no mistake.

Answer and reasoning

The 5 multiplies the logarithm, so divide by 5 first: $\log_3(2x)=2$. Then $2x=3^2=9$, so $x=9/2$. The student's line gives $x=(3^{10})/2$, which is far too large. **The correct solution is $x=9/2$.**

Example 16: Slip 4

$$4+\log_2(3x)=7$$
$$\log_2(3x)=3$$
$$3x=6$$

Question

A student wrote these lines. What is the mistake?

A. Both sides should have been divided by 4, not had 4 subtracted.
B. The base of the logarithm should be 3, not 2.
C. There is no mistake.
D. In the last step, 2 and 3 were multiplied instead of using 3 as an exponent.

Answer and reasoning

Subtracting 4 was right: $\log_2(3x)=3$. But switching forms means the 3 becomes an exponent: $3x=2^3=8$. The student multiplied $2\cdot 3$ instead. So $x=8/3$. Check: $\log_2 8=3$, and $4+3=7$. **The correct solution is $x=8/3$.**